



Systematic Layout Planning (SLP)

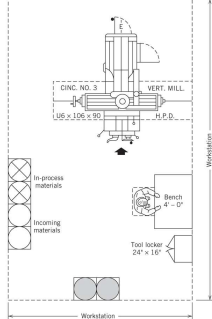
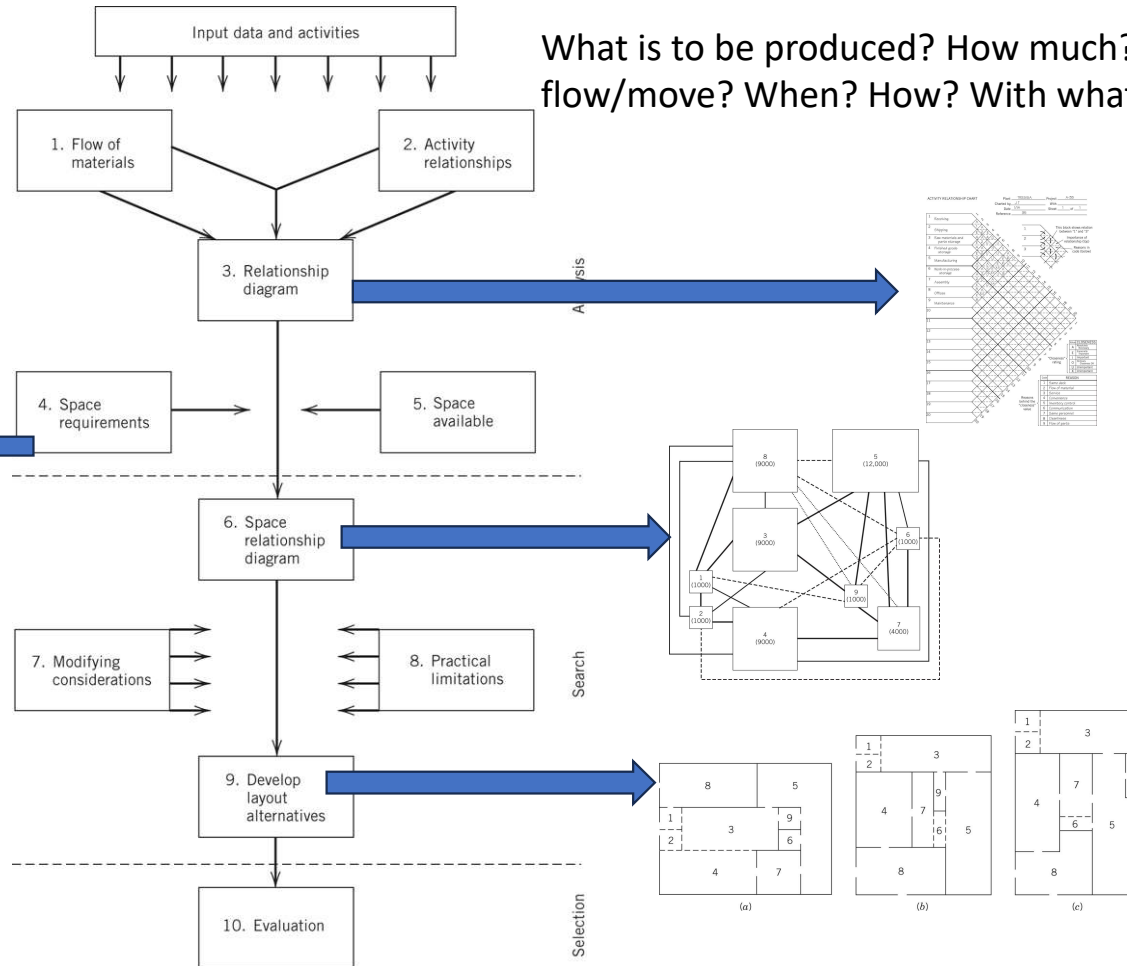
March 2025
W Kruger



SLP diagram logic

What is to be produced? How much? How will it flow/move? When? How? With what? = DATA gathered

Equipment	Equipment area needed	Material area needed	Personnel area needed	Total
Lathes (5)	4x12=48x5=240	20x5=100	20x5=100	240+100+100=440
Screw m/c (6)	4x15=60x6=360	40x6=240	20x6=120	360+240+120=720
Chuckers (2)	6x5=30x2=60	50x2=100	20x2=40	60+100+40=200
Nett total				440+720+200=1360+desk=1364
Aisle %				177
TOTAL AREA				1364+177=1541sq m

Systematic Layout Planning for a Floor of a Department Store – example1 to illustrate SLP logic

A. Relationship Chart (based on Tables B and C)

From	To				Area (sq. ft.)
	2	3	4	5	
1. Credit department	I 6	U —	E 4	U —	100
2. Toy department		U —	I 1	A 1,6	400
3. Wine department			U —	X 1	300
4. Camera department				X 1	100
5. Candy department					100

Letter Closeness rating
Number Reason for rating

1 A Between where is A?
1 E
2 I
4 U
2 X Between where is X's?

B.

CODE	REASON*
1	Type of customer
2	Ease of supervision
3	Common personnel
4	Contact necessary
5	Share same space
6	Psychology

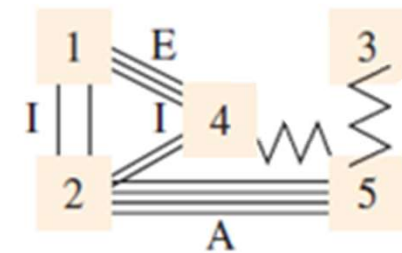
*Others may be used.

C.

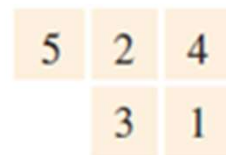
VALUE	CLOSENESS	LINE CODE*	NUMERICAL WEIGHTS
A	Absolutely necessary	=====	16
E	Especially important	=====	8
I	Important	=====	4
O	Ordinary closeness OK	=====	2
U	Unimportant	=====	0
X	Not allowed	^ ^ ^	-80

*Used for example purposes only.

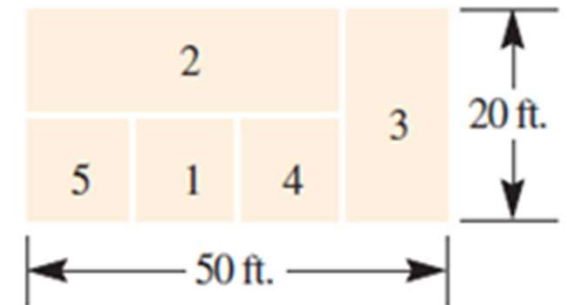
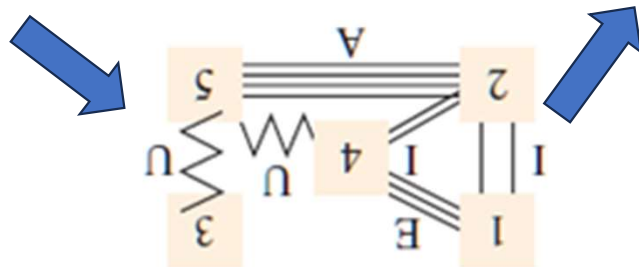
Systematic Layout Planning for a Floor of a Department Store – example to illustrate SLP logic



Initial relationship diagram
(based on Tables A and C)



Initial layout based on
relationship diagram
(ignoring space and
building constraints)



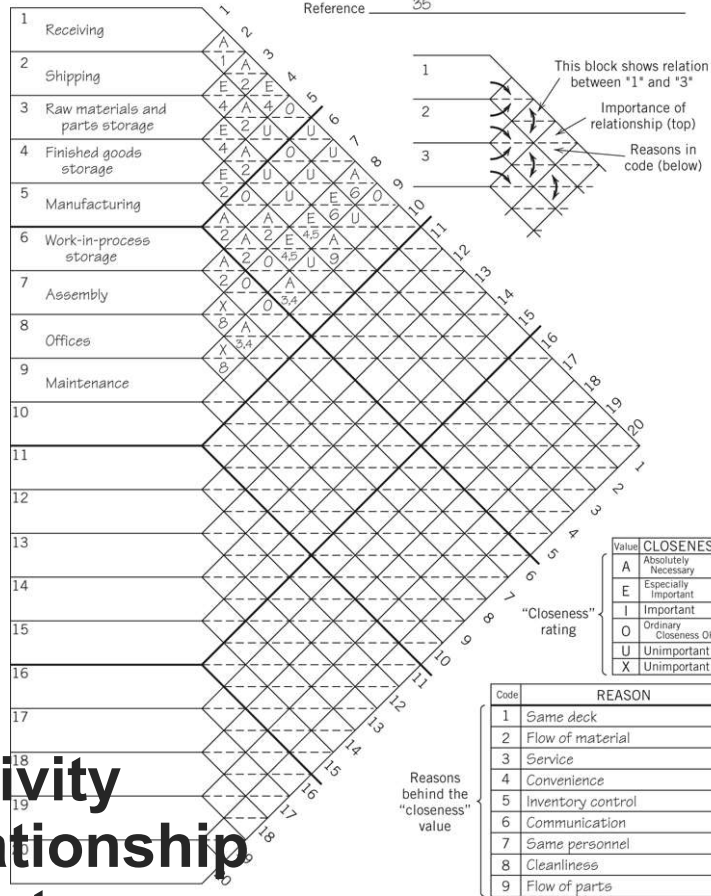
Final layout adjusted by square
footage and building size

SLP – example2 p300 to illustrate SLP logic with activity relationship chart given

Activity relationship chart

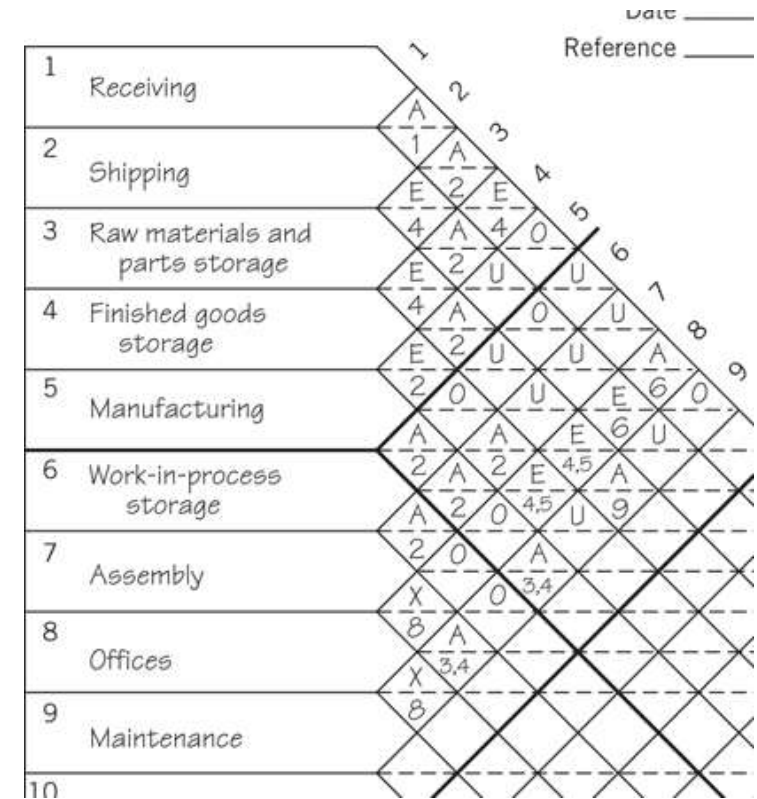
ACTIVITY RELATIONSHIP CHART

Plant TRESISA Project A-35
Charted by J.T With _____
Date 1/14 Sheet 1 of 1
Reference 35

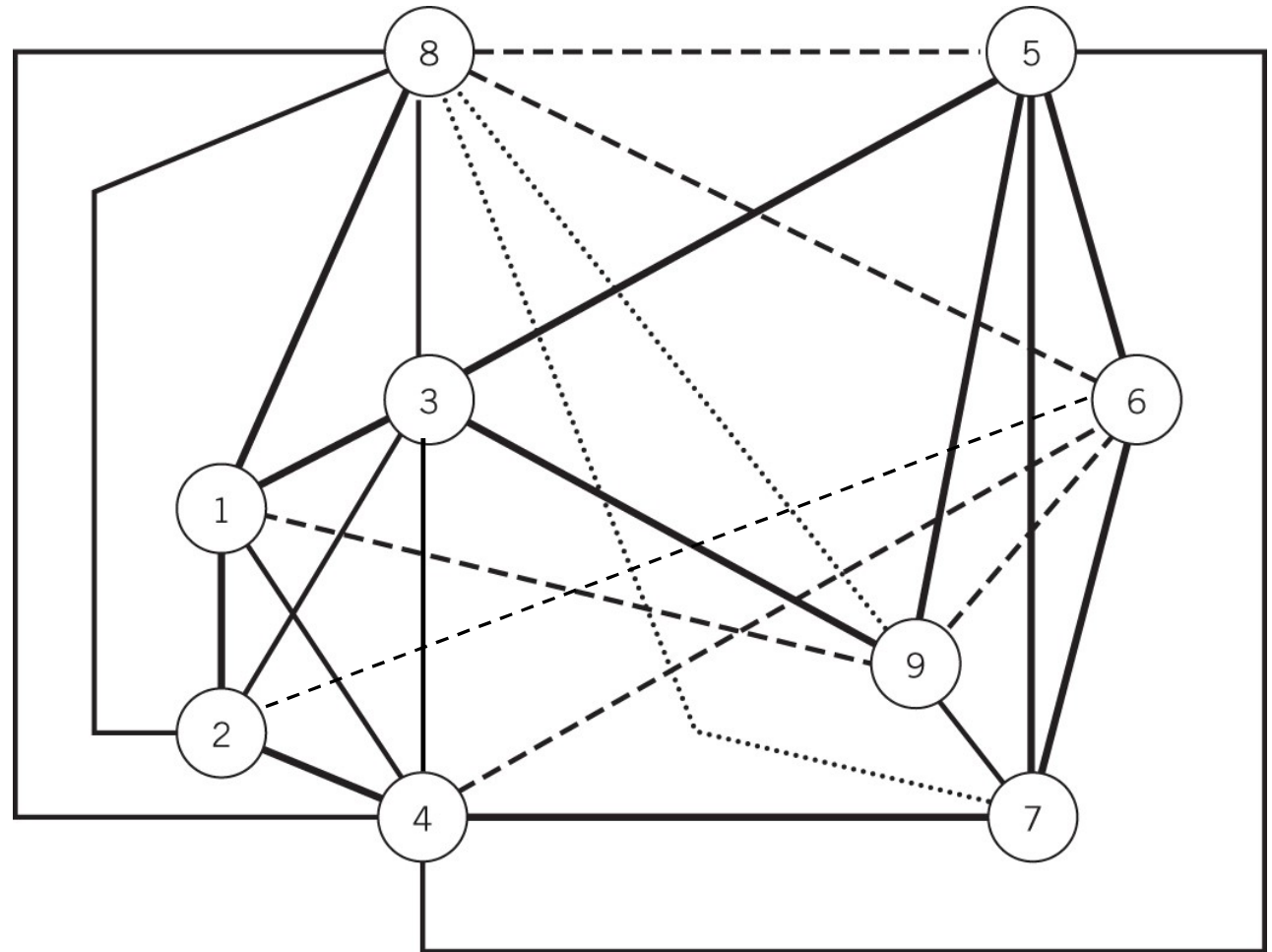


REASONS =
Out of Scope

A – 10 **=====**
E – 7 **=====**
I – 0 **-----**
O – 7 **-----**
U – 8 **no line**
X – 2 **.....**

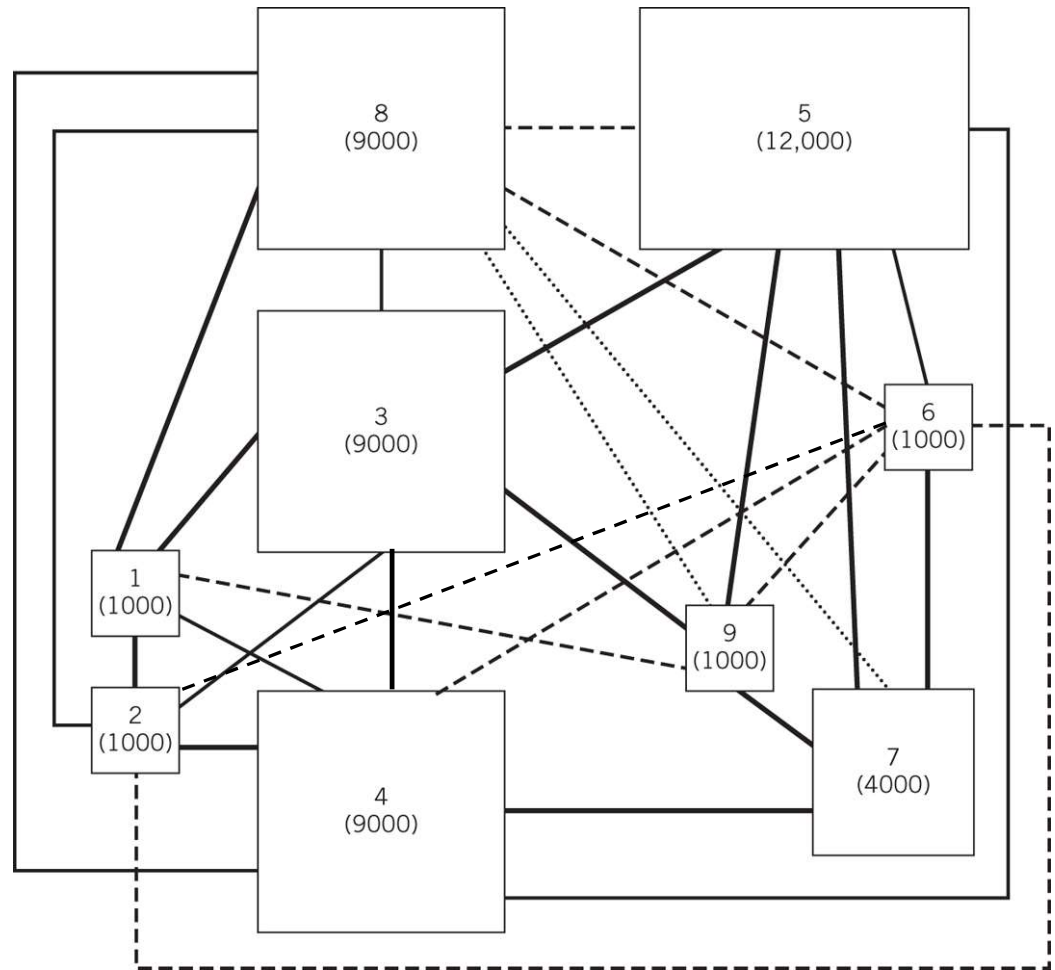


Resulting Relationship diagram

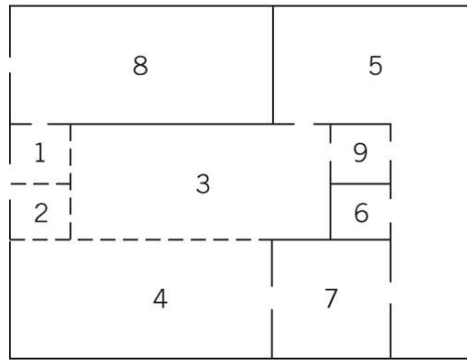


Resulting Space relationship diagram

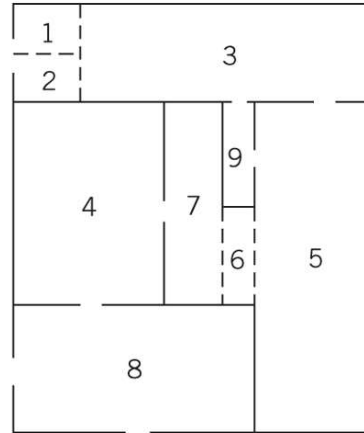
(imagine rubber
bands)



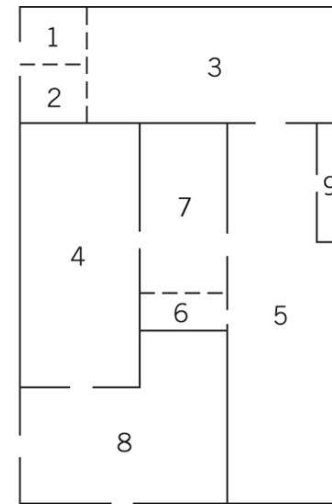
Resulting Alternative Block Layouts



(a)



(b)



(c)

fig_06_07

From-To chart

Department Name	Area (ft ²)	No. of Grids	Flow							
			A	B	C	D	E	F	G	H
1. A: Receiving	12,000	30	0	45	15	25	10	5	0	0
2. B: Milling	8000	20	0	0	0	30	25	15	0	0
3. C: Press	6000	15	0	0	0	0	5	10	0	0
4. D: Screw m/c	12,000	30	0	20	0	0	35	0	0	0
5. E: Assembly	8000	20	0	0	0	0	0	65	35	0
6. F: Plating	12,000	30	0	5	0	0	25	0	65	0
7. G: Shipping	12,000	30	0	0	0	0	0	0	0	0
8. H: Dummy	2000	5	0	0	0	0	0	0	0	0

table_06_03

Ranking (or given, gut feel, or sometimes calculation):

Max = 90 / 5 = intervals

0-18 = U

19-36 = O

37-54 = I

55-72 = E

73-90 = A

SLP – example 6.1 starting from From-To chart Ex 6.1-6.4

Constraints – Receiving(A) and Shipping(G) already placed next to each other, H = expansion space

Activity relationship chart

	A	B	C	D	E	F	G	H		A	B	C	D	E	F	G	H
A	0	45	15	25	10	5	0	0	A	—	I	U	O	U	U	U	U
B		0	0	50	25	20	0	0	B		—	U	I	O	O	U	U
C			0	0	5	10	0	0	C			—	U	U	U	U	U
D				0	35	0	0	0	D				—	O	U	U	U
E					0	90	35	0	E					—	A	O	U
F						0	65	0	F						—	E	U
G							0	0	G							—	U
H								0	H								—

(a) Flow-between chart

(b) Relationship chart

table_06_04

SLP – try to get the Block layout here

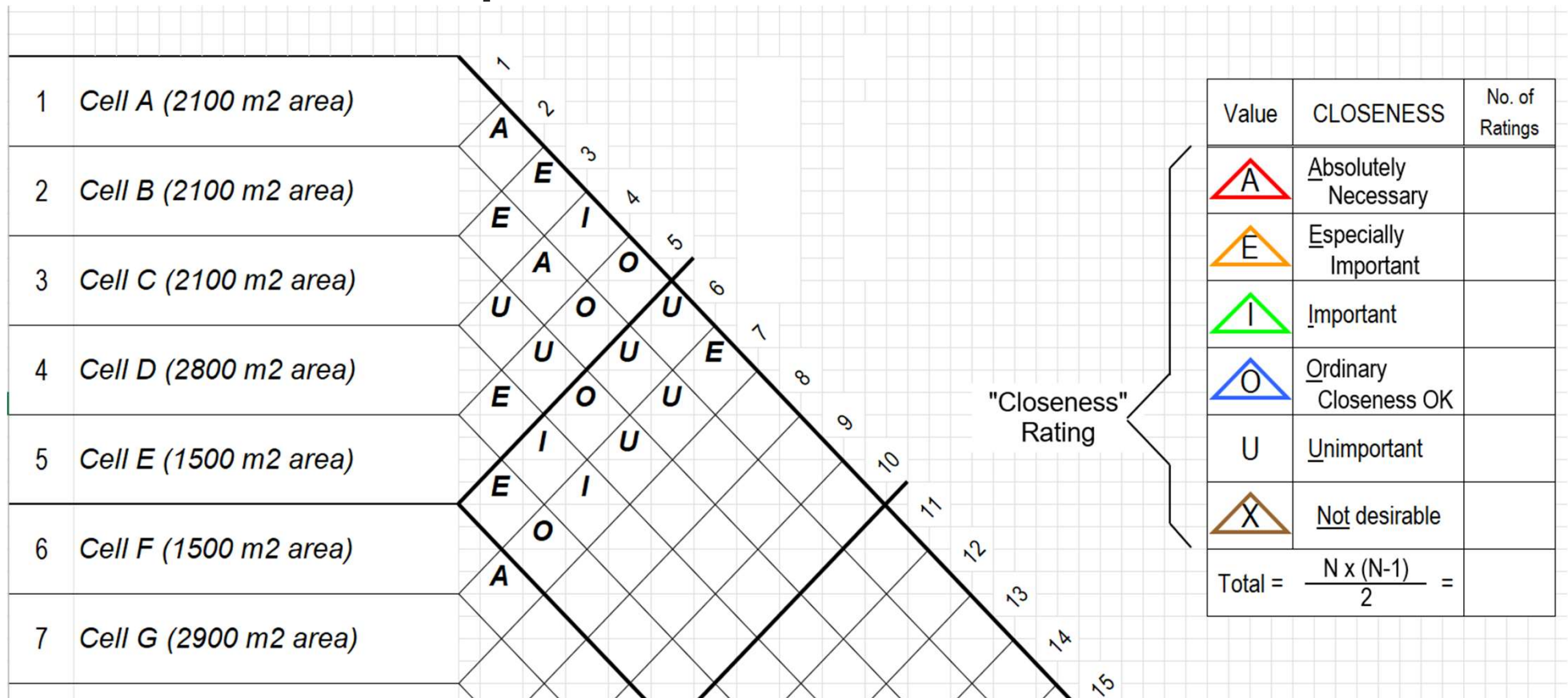
SLP – further examples....

Problems 6.10 in Tompkins book – 7 departments, different sizes

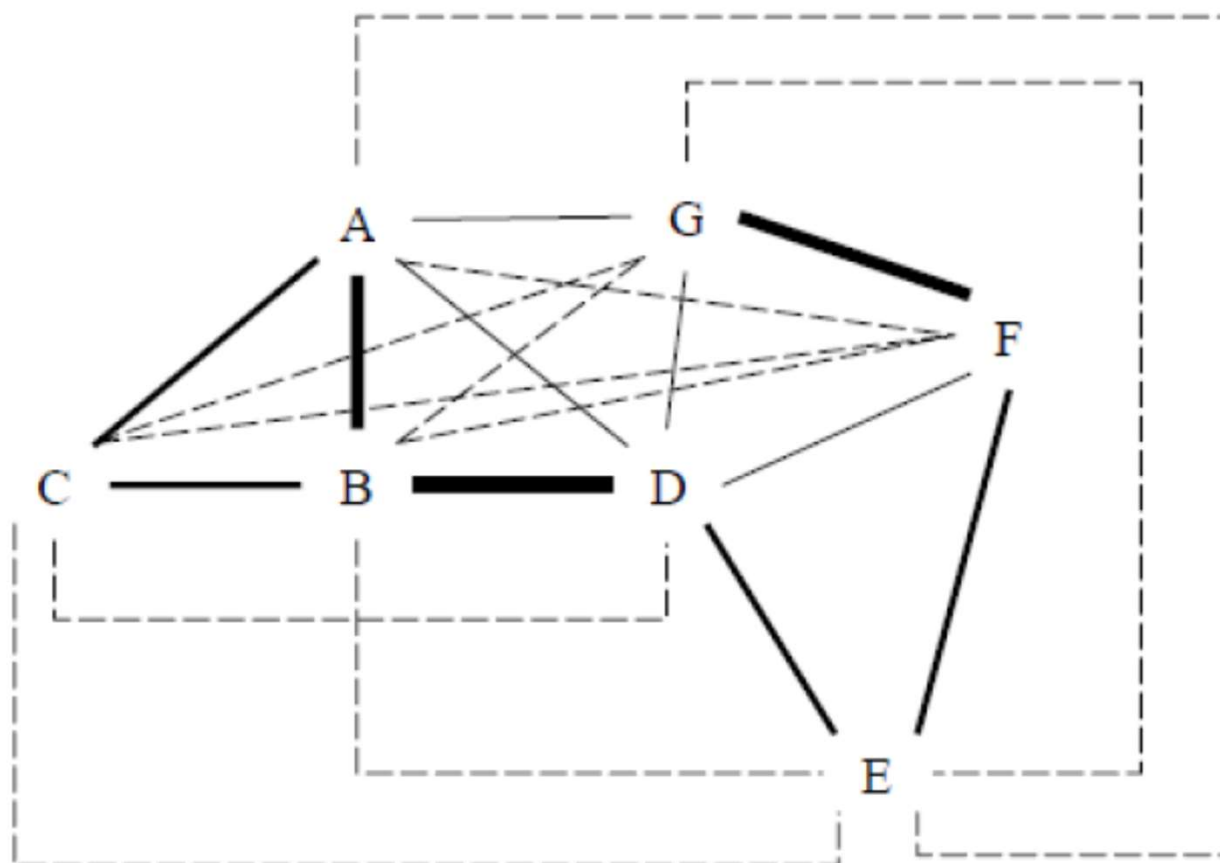
Chase textbook ch8 question 11 (pdf file) – 8 departments, same size, nr of lines is like rubberband

www.richardmuther.com for free working forms.(RMA130)

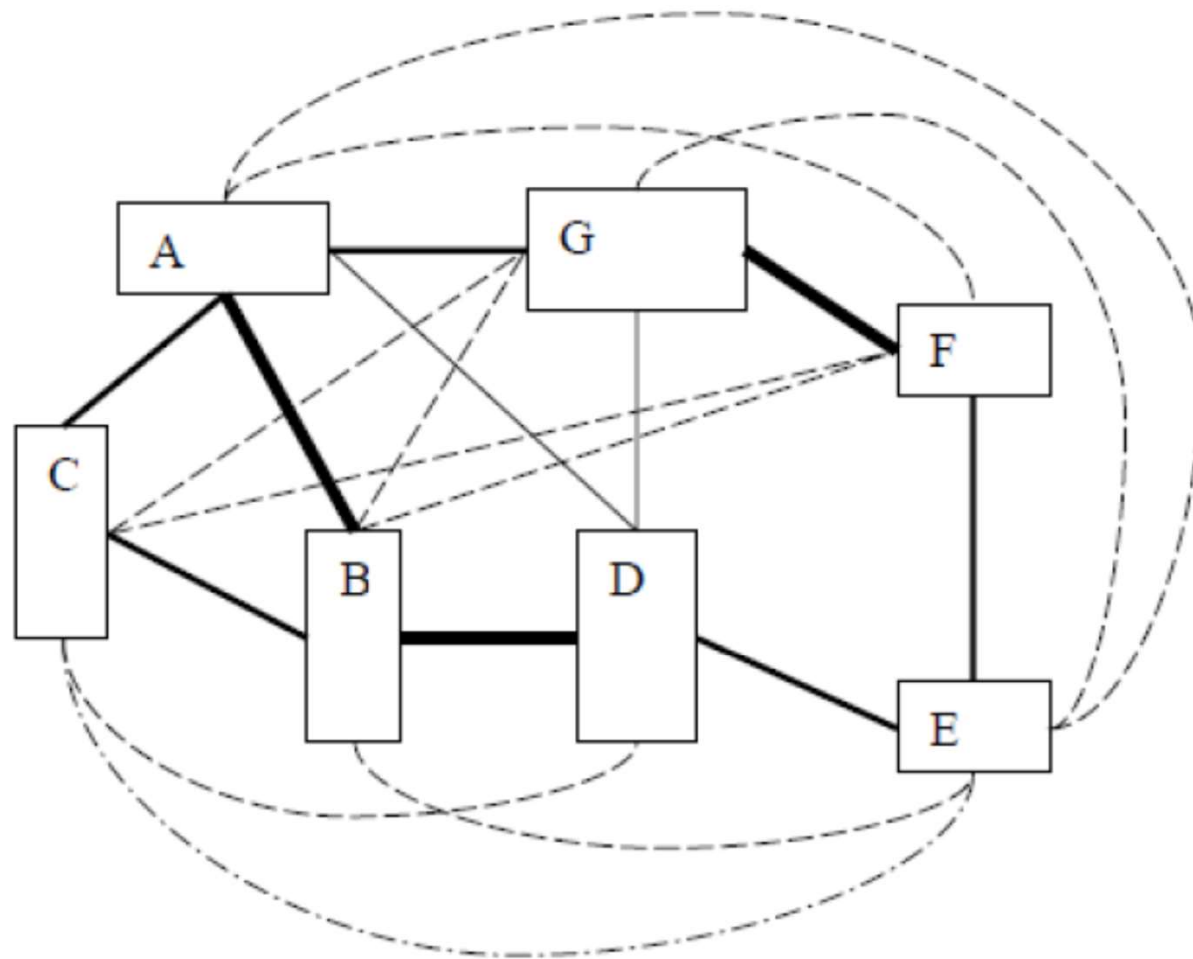
Problem 6.10 – 7 departments



6.10a Relationship Diagram



6.10 Space Relationship Diagram



SLP – try to get a Block layout (or Alternatives) here

TUTORIAL :

6.8 with 6 departments starting with From-To chart – AIM is to get a BLOCK LAYOUT

6.8a From-To Chart

Dept.	A	B	C	D	E	F
A	–	13,320	1,800	0	0	0
B	0	–	11,400	6,600	4,920	5,400
C	0	6,600	–	2,400	4,200	3,600
D	0	3,000	1,200	–	5,040	960
E	0	5,400	7,800	1,200	–	5,160
F	0	0	0	0	0	–

TUTORIAL :

Q11 from Chase book, 8 departments starting with From-To chart – AIM is to get a BLOCK LAYOUT

DEPARTMENTS								
	1	2	3	4	5	6	7	8
1	—	20						
2	15	—	25				4	
3		5	—	40	5			
4			5	—	10			
5	1			20	—	30		
6						—	20	
7				3			—	10
8							5	—

TUTORIAL :

and yet another one

